branching wave function. Happily, for these purposes we can concentrate our attention on the quantum field theory vacuum—the physics of empty space, without any particles flying around.

To bring home the interestingness of the field-theory vacuum, let's focus on one of its most obvious aspects, its energy. It's tempting to think that the energy is zero by definition. But we've been careful not to say that: the vacuum is the "lowest-energy state," not necessarily a "zero-energy state." In fact, its energy can be anything at all; it's a constant of nature, a parameter of the universe that is not determined by any other set of measurable parameters. As far as quantum field theory is concerned, you have to just go out and measure what the energy of the vacuum actually is.

And we have measured the vacuum energy, or at least we think we have. It's not easy to do; you can't simply put a cupful of empty space on a scale and ask how much it weighs. The way to do it is to look for the gravitational influence of the vacuum energy. According to general relativity, energy is the source of the curvature of spacetime, and therefore of gravity. The energy of empty space takes a particular form: there is a precisely constant amount in every cubic centimeter of space, unchanging through the universe, even as spacetime expands or warps. Einstein referred to the vacuum energy as the *cosmological constant*, and cosmologists long debated whether its value was exactly zero or some other number.

That debate seems to have been settled in 1998, when astronomers discovered that the universe is not only expanding but also accelerating. If you look at a distant galaxy and measure the velocity with which it is receding, that velocity is increasing with time. That would be extremely surprising if all the universe contained were ordinary matter and radiation, both of which have the gravitational effect of pulling things together and slowing down the expansion rate. A positive vacuum energy has the opposite effect: it pushes the universe apart, leading to accelerated expansion. Two teams of astronomers measured the distances and velocities of extragalactic supernovae, expecting to measure the deceleration of the universe. What they actually found was that it is speeding up. The discomfiting surprise at obtaining such an unexpected result was partly ameliorated by winning the Nobel Prize in 2011. (The debate "seems to" have been settled, because it's still an open possibility that cosmic acceleration is caused by something other than vacuum energy. But that's